WORKED SOLUTION

STEP III 2015, Question 6

Pure • Symmetric Functions of Roots

STEP III 2015, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s3-q6

The key idea

Everything about the pair w, z is controlled by its sum and its product, because w and z are the two roots of $t^2 - (w+z)t + wz = 0$. Part (i) is then a statement about when a quadratic with real coefficients has real roots, and the condition $u^2 \leq 2v$ is the discriminant in disguise. In part (ii) the first two equations fix the product wz, and the third then becomes a cubic in u alone, with a root you can spot.

Solution

Part (i)

The identity that drives everything is

$$2v - u^2 = 2(w^2 + z^2) - (w+z)^2 = w^2 - 2wz + z^2 = (w-z)^2.$$

We also note that $wz = \frac{1}{2}((w+z)^2 - (w^2+z^2)) = \frac{1}{2}(u^2 - v)$.

If w and z are real: then $u = w+z$ and $v = w^2+z^2$ are certainly real, and $2v - u^2 = (w-z)^2 \geq 0$ because $w - z$ is real. So $u^2 \leq 2v$.

Conversely, suppose u and v are real and $u^2 \leq 2v$. Then $p = wz = \frac{1}{2}(u^2 - v)$ is real too. Since

$$(t-w)(t-z) = t^2 - ut + p,$$

the numbers w and z are precisely the two roots of the quadratic $t^2 - ut + p = 0$, whose coefficients are real. Its discriminant is

$$u^2 - 4p = u^2 - 2(u^2 - v) = 2v - u^2 \geq 0,$$

so both roots are real. Hence w and z are real, which completes the proof of the equivalence.

? **Ask yourself** Why isn't "u and v real" enough on its own?

Because a conjugate pair also has real sum and real sum of squares. For example $w = 1 + 2\mathrm{i}$, $z = 1 - 2\mathrm{i}$ give $u = 2$ and $v = -6$, both real, but $u^2 = 4 > -12 = 2v$. The inequality is exactly what rules out a non-real conjugate pair.

Part (ii)

Reducing to a cubic. The first equation says $u = w + z$. Writing $v = w^2 + z^2$, the second says $v = u^2 - \frac{2}{3}$, so

$$wz = \tfrac{1}{2}(u^2 - v) = \tfrac{1}{3}.$$

For the cubes we use the standard identity

$$w^3 + z^3 = (w + z)^3 - 3wz(w + z) = u^3 - u.$$

The third equation therefore becomes $u^3 - u - \lambda u = -\lambda$, that is

$$u^3 - (1 + \lambda)u + \lambda = 0.$$

Substituting $u = 1$ gives $1 - 1 - \lambda + \lambda = 0$, so $u - 1$ is a factor, and

$$u^3 - (1 + \lambda)u + \lambda = (u - 1)(u^2 + u - \lambda).$$

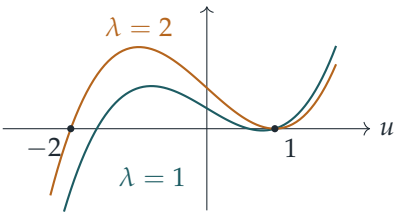
Counting the roots. The quadratic factor has discriminant $1 + 4\lambda > 0$, since $\lambda > 0$. So it has two distinct real roots,

$$u = \frac{-1 \pm \sqrt{1 + 4\lambda}}{2}.$$

One of these equals 1 exactly when $1^2 + 1 - \lambda = 0$, that is when $\lambda = 2$. So:

- if $\lambda \neq 2$, the possible values of u are 1 and $\frac{1}{2}\left(-1 \pm \sqrt{1 + 4\lambda}\right)$: three distinct values, all real;
- if $\lambda = 2$, the cubic is $(u - 1)^2(u + 2)$ and there are only two values, $u = 1$ and $u = -2$.

Each of these values really does occur: given u, take w and z to be the roots of $t^2 - ut + \frac{1}{3} = 0$. Then $w + z = u$ and $wz = \frac{1}{3}$, and reversing the steps above shows that all three equations hold. So the exceptional value is $\lambda = 2$.



The sketch (vertically scaled) shows $u^3 - (1 + \lambda)u + \lambda$ for $\lambda = 1$, with three separate crossings, and for $\lambda = 2$, where the curve just touches the axis at $u = 1$.

Are w and z necessarily real? No. Since u is real and $v = u^2 - \frac{2}{3}$ is real, part (i) tells us that w and z are real if and only if $u^2 \leq 2v = 2u^2 - \frac{4}{3}$, that is, if and only if $u^2 \geq \frac{4}{3}$. The root $u = 1$ is present for every λ, and $1 < \frac{4}{3}$, so it always gives non-real w and z.

Explicitly, $u = 1$ gives $t^2 - t + \frac{1}{3} = 0$, so

$$w = \frac{1}{2} + \frac{\mathrm{i}}{2\sqrt{3}}, \qquad z = \frac{1}{2} - \frac{\mathrm{i}}{2\sqrt{3}}.$$

As a check: $w + z = 1 = u$; $wz = \frac{1}{4} + \frac{1}{12} = \frac{1}{3}$, so $w^2 + z^2 = 1 - \frac{2}{3} = u^2 - \frac{2}{3}$; and $w^3 + z^3 = u^3 - 3wz\,u = 1 - 1 = 0$, so $w^3 + z^3 - \lambda u = -\lambda$ for every $\lambda > 0$. This is a counterexample: the three values of u are all real, but w and z need not be.

Checks

- With $\lambda = 1$ the values of u are 1 and $\frac{1}{2}(-1 \pm \sqrt{5}) \approx 0.618$, -1.618, and each satisfies $u^3 - 2u + 1 = 0$.
- With $\lambda = 2$ the cubic $u^3 - 3u + 2$ vanishes at $u = -2$ ($-8 + 6 + 2 = 0$), and both it and its derivative $3u^2 - 3$ vanish at $u = 1$, confirming the double root.
- A computer check over many values of λ built w and z from each u and confirmed all three equations to rounding error; it also found three distinct values of u except at $\lambda = 2$.

Common slips

- In part (i), proving only one direction. The converse needs the product wz to be real, which is where the reality of v is used.
- Forgetting to check that the quadratic's roots are different from 1, and so missing the exceptional value $\lambda = 2$.
- Assuming that because u is real, w and z must be. Real symmetric functions of w and z are equally consistent with a conjugate pair.

Beyond STEP

The idea that a pair (or any set) of numbers is determined by its symmetric functions is the starting point of the theory of symmetric polynomials: every symmetric polynomial in w and z can be written in terms of $w + z$ and wz, and Newton's identities relate power sums like $w^3 + z^3$ to them. The same reasoning, with discriminants, decides when all roots of a real polynomial are real. Needham's *Visual Complex Analysis* is a beautiful companion for complex numbers, and Allenby's *Rings, Fields and Groups* takes symmetric polynomials further.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.